Vulnerability Assessment Report

13th October 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

The database server stores and manages the customer, campaign and analytics data that the marketing team uses to follow results and personalize its work. It is in daily use, so the business depends on it.

- *Securing that data matters because the permissions on the system are very open at the moment. A leak or a change made by the wrong person would hurt customers, could bring fines, and would damage the company's reputation.*
- *If the server were disabled, the marketing operations would stop until it came back online, and the team would lose time and money.*
- *How might the server impact the business if it were disabled?*

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtains sensitive information by exfiltration, helped by the open access permissions	3	3	9
Employee	Disrupts mission-critical operations, or copies data using more access than needed	2	3	6
Customer	Alters or deletes critical information	1	3	3

Hacker	Reads data in transit because the connection uses old SSL	2	2	4

Approach

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database.